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Data-driven Design of Electrolyte Additives for High-Performance 5 V LiNi0.5Mn1.5O4 Cathodes

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Article

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Abstract

$LiNi_{0.5}Mn_{1.5}O_4$ (LNMO) is a high-capacity spinel-structured material with an average lithiation/de-lithiation potential at ca. 4.6–4.7 V, far exceeding the stability limits of electrolytes. An efficient way to enable LNMO in lithium-ion batteries is to reformulate an electrolyte composition that stabilizes both graphitic (Gr) anode with solid-electrolyte-interphase (SEI) and LNMO with cathode-electrolyte-interphase (CEI). In this study, we selected and tested a diverse collection of 28 single and dual additives for the LNMO||Gr system. Subsequently, we trained machine learning (ML) models using this dataset and employed these models to identify 6 optimal binary compositions out of 125, based on their predicted final area-specific-impedance, impedance-rise, and final specific-capacity. The additives generated through this ML approach demonstrated superior performance compared to those in the in the initial dataset. This finding not only underscores the efficacy of ML in identifying new materials in a highly complicated application space, but also showcases an accelerated material discovery workflow that directly integrates data-driven methods with battery testing experiments.

1. Introduction

The spinel-structured $LiNi_{0.5}Mn_{1.5}O_4$ exhibits an average lithiation/de-lithiation voltage at ~ 4.7 V with a high specific capacity and excellent rate capability, making it a promising candidate as cathode material for high-energy lithium-ion batteries (LIBs), while the absence of cobalt (Co) in it brings additional advantage considering the geopolitical as well as ethical risks associated with mining of Co. However, serious challenges also arise from the high operating voltage of LNMO, which far exceeds the stability limit of any known electrolyte. One typical example of the reaction between electrolyte and LNMO is the oxidative decomposition of ethylene carbonate (EC), a prevalent electrolyte solvent in mainstream LIB industry, which forms glycolic acid and difluorophosphoric acid (HPO_2F_2) accompanied with the reduction of transition metal cores and their concomitant dissolution. The dissolved species such as Mn(II) further engage in the cross-talk between cathode and anode, where it deposits on anode surface in either metallic or ionic form, resulting in additional capacity loss as well as cell impedance rise.[1, 2] To make LNMO chemistry reversible, utilization of the electrolyte additives is the most efficient and economical approach, which, without significantly changing in the mainstream electrolyte formulation and supply chain, offers several advantages. Among them are low cost, direct interphasial engineering, and minimized side effects on other important properties of electrolytes such as ion transport, chemical compatibility with other cell parts, as well as the viscosity and rheology that are already integrated as part of the mature LIB manufacturing protocol. However, the massive chemical space of electrolyte additives together with long cycling experiments often render any large-scale screening effort practically impossible.

Machine learning (ML) has rapidly become a new paradigm in the field of materials science, offering unprecedented acceleration in materials discovery and optimization.[3] ML techniques enable the prediction of material properties, design of material structures with desired functionality, and

identification of novel material candidates through the analysis of extensive and multifaceted datasets.[4] This approach has significantly reduced the time and cost associated with traditional experimental approaches, particularly in domains critical to technological progress such as energy storage[5] and catalysis.[6] In the realm of battery technology, the impact of ML has been profound for accelerated screening of liquid[7, 8, 9] and solid electrolytes.[10, 11, 12] Specifically, ML algorithms have been used to predict the redox potentials of electrolyte additives,[13, 14] as well as coulombic efficiency[15] and cycle life[16] of LIBs as a function of additive formulas, facilitating the identification of compounds that could lead to superior battery performance.

In this study, we proposed employing machine learning predictions of key battery's performance metrics to accelerate the discovery of optimal electrolyte additives. The overall ML-guided experimental workflow is illustrated in Fig. 1a, and stepwise ML tasks are shown in Fig. 1b. We initially compiled a diverse collection of electrolyte additives and examined their impact on the performance of LNMO||Gr cells, accompanied by extensive characterizations (Fig. 1b, Step 1). This dataset allowed us to explore the structure-property relationship between additives and three key performance indicators of the battery, namely the final area specific impedance (ASI), impedance rise (ΔASI), and final specific capacity (Q). Among these, lower ASI and ΔASI indicate higher power density, improved charging/discharging rate, and enhanced efficiency, whereas a higher Q represents higher energy density for a given weight of the battery. We then trained and evaluated ML models with the dataset so that they can predict ASI, ΔASI, and Q (Fig. 1b, Step 3) based on the chemical formulas and compositions of additives. Finally, we applied the trained models on an unknown set of 125 dual additives, of which the predicted metrics were utilized to determine the most promising candidates for experimental validation (Fig. 1b, Step 4 &5).

2. Result and Discussion

2.1. Electrochemical Performance

Our approach started with collecting a diverse set of additives that have been reported in the literature. These additives have been shown to contribute to the improved performance of either cathode or anode by reducing impedance, preventing lithium inventory loss, and mitigating electrolyte hydrolysis. In this paper, the beneficial additives for cathodes are referred to as cathode additives, while those benefitting anode are referred to as anode additives. The baseline solvent is 1.0 M $LiPF_6$ in EC/EMC at 1/9 volumetric ratio, whose performance will be used as a reference. In our list, there are 14 cathode additives and 10 anode additives (Fig. 2). The most commonly used cathode additives include lithium difluorophosphate (LDF), [17] *in situ* generated lithium malonato tetrafluorophosphates (MS),[18] and aged trimethylsilyl phosphite (TMSPi) (Scheme S1).[19] Similarly, anode additives comprises of several typical choices including lithium difluorooxalato borate (LiDFOB),[20] vinylene carbonate (VC),[21] phenylboronic acid 1,3-propanediol ester (PBE),[22] trivinylcyclotriboroxane pyridine complex (tVCBO), etc.[23] Overall, their chemical structures consist of up to seven different elements, namely C, H, Li, P, F, O, and Si. In addition,

various functional groups are present in these additives, including phenyl (C_6H_5), phosphine oxide (X_3P = O), P-F, malonato (-O-C(= O)-CH_2-C(= O)-O-), trimethylsilyl (-Si(OCH_3)$_3$), carboxyl (-C(= O)-O-), B-F, B-O, B-C, and alkene (-C = C-).

From our collection of anode and cathode additives, we further curated and tested 10 single and 18 dual additive systems of various weight percentages (wt%). In this work, the dual additives always consist of a cathode and an anode additive each, as we hypothesized that their co-existence in the electrolyte and the synergistic effects would be critical to stabilize the two electrodes at their respective extreme potentials simultaneously. The distributions of ASI, ΔASI, and Q corresponding to 28 additives, as well as the baseline electrolyte solvent are shown in Fig. 3 (tabulated data in Table S1). In general, these distributions were found to have non-normal trends, skewing either to the left (ASI and ΔASI) or to the right (Q) of their respective range of values. Although multiple additives contribute to improvement over the baseline in one or two performance metrics, only two dual additives, specifically tVCBO at 0.25 wt% and MS at 1.0 wt%, and LiDFOB at 1.0 wt% and TMSPi at 1.0 wt%, surpass the baseline across all three evaluated metrics, achieving lower ASI and ΔASI, as well as a higher specific capacity. It is also noted that many additives containing tVCBO or LiDFOB show enhanced capacity retention compared to the baseline system (Fig. S2)

2.2. Structure-property relationship

Identifying the structure-property relationships of additives is critical for identifying the impacts (whether positive or negative) of structural features/descriptors on certain targeted properties. The assignment of the descriptors/features for additives here is inspired by the previous work of Okamoto et al,[14] wherein the frequency/count of each atom and its coordination in the structure was tabulated. To further distinguish atoms beyond their coordinations, we also incorporated additional physicochemical properties such as formal charge and whether the atom is part of a ring (Fig. 4a). For example, the feature **B[-1]_4_inRing** can be expalined as follows: B represents the element Boron, [-1] indicates the formal charge of -1, the number 4 after the underscore is the coordination (number of neighboring atoms except for H), and “inRing” indicates that the atom B is part of a ring. Note that the descriptor values, or the counts of distinct atomic features, were normalized to account for various concentrations of the additives. A full list of generated features and their calcuated values corresponding to 28 electrolyte addtives in the initial dataset is provided in the SI (Spreadsheet titled “Feature table for SI.xlsx”).

By analyzing the correlations between descriptors and performance metrics, we can extract the influence of each descriptor systematically. In this work, we utilized Spearman correlation analysis which describes how well the relationship between feature and performance metric can be describe as a monotonic function. The most relevant features, based on Spearman correlation cofficients, with respect to ASI, ΔASI, and Q are shown in Fig. 4b, 4c, and 4d, respectively. In these plots, positive and negative monotonic trend between each feature and the performance metrics are indicated by positive and negative values. Notably, among the most negatively correlated features (Spearman correlation coefficient < -0.2) of additives with respect to impedance include B[-1]_4_inRing, P[-1]_6_inRing, Si_4, and

N_2_inRing, respectively. Indeed, the additive combination of 1% LiDFOB and 1% MS (Fig. 4a), where both B[-1]_4_inRing and P[-1]_6_inRing features are present, has the lowest measured impedance (44.21 Ωcm^2). Furthermore, these findings align remarkably well with our current knowledge of addive effects on battery performance: 1) B[-1]_4_inRing implies that the chemical structures of lithium bisoxolatoborate (LiBOB) and lithium difluorobisoxolatoborate (LiDFOB) serve as beneficial cathode electrolyte interphase (CEI) agents.[33] 2) P[-1]_6_inRing suggests that oxyfluorophosphate-based cathodes are favorable for low resistance and robust CEI formation; [34, 35] 3) Si_4 indicates that the presence of a scavenging group, such as trimethylsilyl, effectively reduces impedance;[34, 36] 4) N_2_inRing suggests that a basic group like pyrrole or morpholine behaves as an HF scavenger, reducing transition metal (TM) dissolution.[34, 35] These empirical results further reinforce our design principles for cathode additive, demonstrating the consistency between the observed correlations in this work and the previous research findings.

As illustrated in Fig. 4c, a similar trend was observed in the Spearman correlation of the descriptors with impedance rise, with slight diferrence observed in some new features of P[-1]_4, and O_2_inRing. The P[-1]_4 feature is associated with oxyfluorophosphate such as $LiPO_2F_2$ and $LiPO_3F$, while O_2_inRing in this case is associated with boroxane structure such as in tVCBO and PBE. The correlation between the features and final specific capacity (Fig. 4d) is less insightful, as it is influenced by vairous interplaying factors of transition metal dissolution, lithium inventories loss, impedance, and SEI robustness. Nevertheless, we still can obtain some general features that carry certain chemistry significance, for example, the features that are most positively related to final specific capacity coincide with those that are inversely related to final impedance, such as P[-1]_6_inRing, B[-1]_4_inRing, and N_2_inRing. This suggests that these features are desired as they lead to both specific capacity improvement and reduction in impedance rise.

2.3. Machine Learning Models

To accelerate the search for new additives, it is essential to develop predictive capability ahead of tedious experiments, which typically require several months to complete. Hence, we utilized the above initial dataset to train ML models to predict potential chemical structures and compositions that could lead to improvements in ASI, ΔASI, and Q metrics. Specifically, Gaussian Process Regression (GPR) is the ML model of choice as it has been shown to be one of the most reliable algorithms for low-dimensional and small datasets,[37] which is the case in this work. In addition, GPR also produces uncertainty quantification for every prediction, allowing for quality evaluation of the prediction (see Machine Learning methods in the SI). To further enhance the assessment of our models' reliability, we implemented 10-fold cross-validation (CV), in which the dataset is partitioned into 10 equal segments. During each iteration, one segment is reserved for testing while the remaining nine are utilized for training. This procedure is conducted ten times, with each iteration featuring a distinct test set. The overall error is determined by averaging the errors across all ten models. For all models, mean absolute error (MAE) is employed as the evaluation metric. The parity plots comparing GPR predictions with experimental measurements of 28 additives and the baseline systems are shown in Fig. 5. Based on the

results, the highest prediction accuracy is observed for final specific capacity model (Test $MAE^{10-fold\ CV}$ = 10.7 ± 4.6 $mAhg^{-1}$), followed by impedance rise model (Test $MAE^{10-fold\ CV}$ = 15.3 ± 7.8 Ωcm^2) and final area specific impedance model (Test $MAE^{10-fold\ CV}$ = 20.0 ± 10.6 Ωcm^2). Overall, we believe that our ML models are reasonably accurate given the size of the current training dataset.

2.4. Prediction and validation

To identify new additives with improved performance, we systematically examined every possible combination of dual additives, totaling 140 pairs, by mixing 14 cathode and 10 anode additives in equal weight percentages of 1%. Among these, 15 have already been tested and included in the initial dataset, which leaves 125 additive combinations yet to be explored. Using our trained GPR models, we performed prediction of ASI, ΔASI, and Q for 125 unknown additive candidates. The results were tabulated in as shown the SI (Spreadsheet titled “LNMO_wSingle_additive_customFeats_Highlighted.xlxs”). Furthermore, as the accuracy of GPR models has been shown to be more reliable for the prediction of ΔASI and Q with lower MAEs and uncertainty (Fig. 5), we employed those as the ranking criteria for selecting new additive combinations for experimental validation. The experimental measurements for the top 6 dual additives candidates are reported in Table 1, where we identified three out of six dual additives with desirable measured performance metrics (No. 29, 31, and 32). Among these, the dual system comprising of LiDFOB at 1.0 wt% and SA at 1.0 wt% shows similar Q but improved (lower) ASI and ΔASI compared to the baseline solvent. More importantly, the addition of either MS or SA to LiBOB show notable enhancement in all three considered metrics, with the combination of LiBOB at 1.0 wt% and SA at 1.0 wt% achieving the highest final specific capacity (95.49 $mAhg^{-1}$) among all additives in this work.

To gain further insights into additive performance, we carried out an array of experimental and post-test analysis of the cycled cells of the top four additive compositions and the baseline in this study, particularly focusing on the degradation mechanisms, including the regular checkup on the cycled anodes for the TM cross walked from the cathodes (see SI for details). ^1H nuclear magnetic resonance spectroscopy (NMR) clearly shows the inhibition of transesterification in presence of the designed additive combinations (Fig. S8). X-ray photoelectron spectroscopy (XPS) confirms the formation of oxyfluorophosphates in some additives that improves CEI (Fig. S9). Inductive coupled plasma mass spectrometry (ICP-MS) confirms the beneficial effects of additives in reducing transition metal dissolution/deposition on the anode side (Fig. S10). SEM confirms the presence of TM aggregates in cells with additives, thereby reducing their detrimental effects on SEI rejuvenation and lithium inventory consumption (Fig. S11 – S15). The online electrochemical mass spectrometry (OEMS) experiments have shown that these additives can also inhibit the consecutive breakdown and reformation of SEI, a process that leads to lithium inventory consumption (Fig. S16). The experiments on harvested cell components clearly identify the lithium inventory loss as the main degradation mechanism, associated with TM dissolution and high impedance rise (Table S2 and Fig. S7). All these point to the effective mitigation of degradation by these ML-predicted additive formulations.

Table 1
List of machine learning-suggested additives and their measured performance metrics. The baseline solvent (1.0 M $LiPF_6$ in EC/EMC at 1/9 volumetric ratio) is included for reference.

No.	Additive	ASI (Ωcm^2)^	ΔASI (Ωcm^2)*	Q ($mAhg^{-1}$)%
29	LiDFOB@1.0wt%+SA@1.0wt%	41.19	16.26	91.79
30	LiBOB@1.0wt%+TMSPi@1.0wt%	104.76	65.57	69.80
31	LiBOB@1.0wt%+MS@1.0wt%	50.13	16.77	92.17
32	LiBOB@1.0wt%+SA@1.0wt%	46.79	10.19	95.49
32	LiBOB@1.0wt%+SA@1.0wt%	123.53	85.94	47.07
33	LiBOB@1.0wt%+LDF@1.0wt%	203.43	110.36	80.06
	Baseline	54.47	19.35	91.94

^: Area specific impedance; *: Impedance rise; %: Final specific capacity

3. Conclusion

In summary, we successfully showcased a data-driven experimental framework aimed at fast and efficient identification of electrolyte additives for LIBs based on LNMO cathodes. This method utilized a limited set of initial experimental data to develop reliable machine learning models that directed subsequent experimental efforts. We began by creating an initial dataset from performance metrics including final area specific impedance, impedance rise, and final specific capacity, gathered from 28 additives and the baseline solvent. Utilizing this data, we employed ML models to evaluate these performance metrics for an expanded, untested group of 125 dual electrolyte additives. Remarkably, by experimentally validating only the top 6 candidates identified through ML predictions, we discovered a new binary formulation, namely LiBOB at 1wt% and SA at 1wt%, that outperformed all additives in the initial dataset. Future research will explore a broader array of additives, including ternary compositions, via closed-loop experiments guided by Bayesian optimization. The methodology described herein has the potential to be applied universally to other areas of materials discovery, particularly where navigating vast design spaces and conducting time-intensive experiments are major hurdles.

Declarations

Authorship contribution statement

Bingning Wang: Experiment design, Data collection and analysis, Writing - review & editing. **Hieu A. Doan**: Data analysis, Machine learning development, Writing original draft. **Seoung-bum Son**: Data analysis, Writing - review & editing; **Stephen E. Trask**: Experiment design, Writing - review & editing; **Daniel Abraham**: Writing - review & editing; **Andrew Jansen**: Funding, Writing - review & editing; **Kang Xu**:

Supervision,Writing - review & editing; **Chen Liao**: Supervision, Conceptualization, Funding, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figures

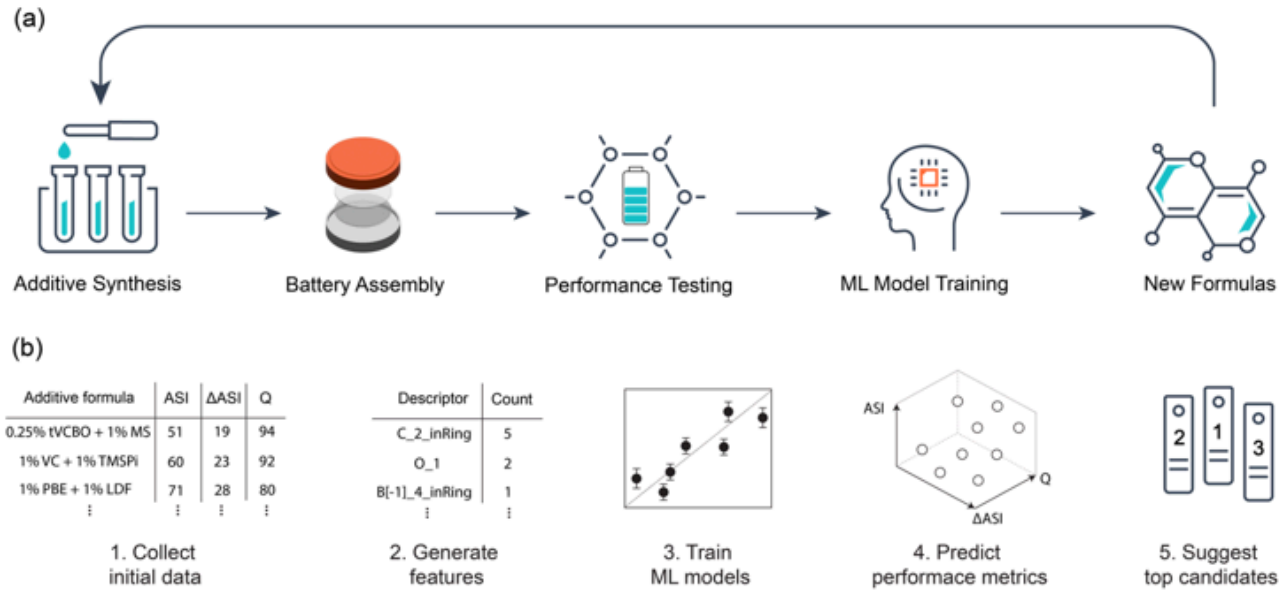

Figure 1

A data-driven experimental workflow for additive discovery and optimization for lithium-ion batteries. (a) Schematic representation of a machine learning (ML)-guided design of experiment workflow for electrolyte additive discovery. (b) Sequential method for developing ML models to predict and recommend new electrolyte additives. ASI, DASI, and Q denote final area specific impedance, impedance rise, and final specific capacity, respectively.

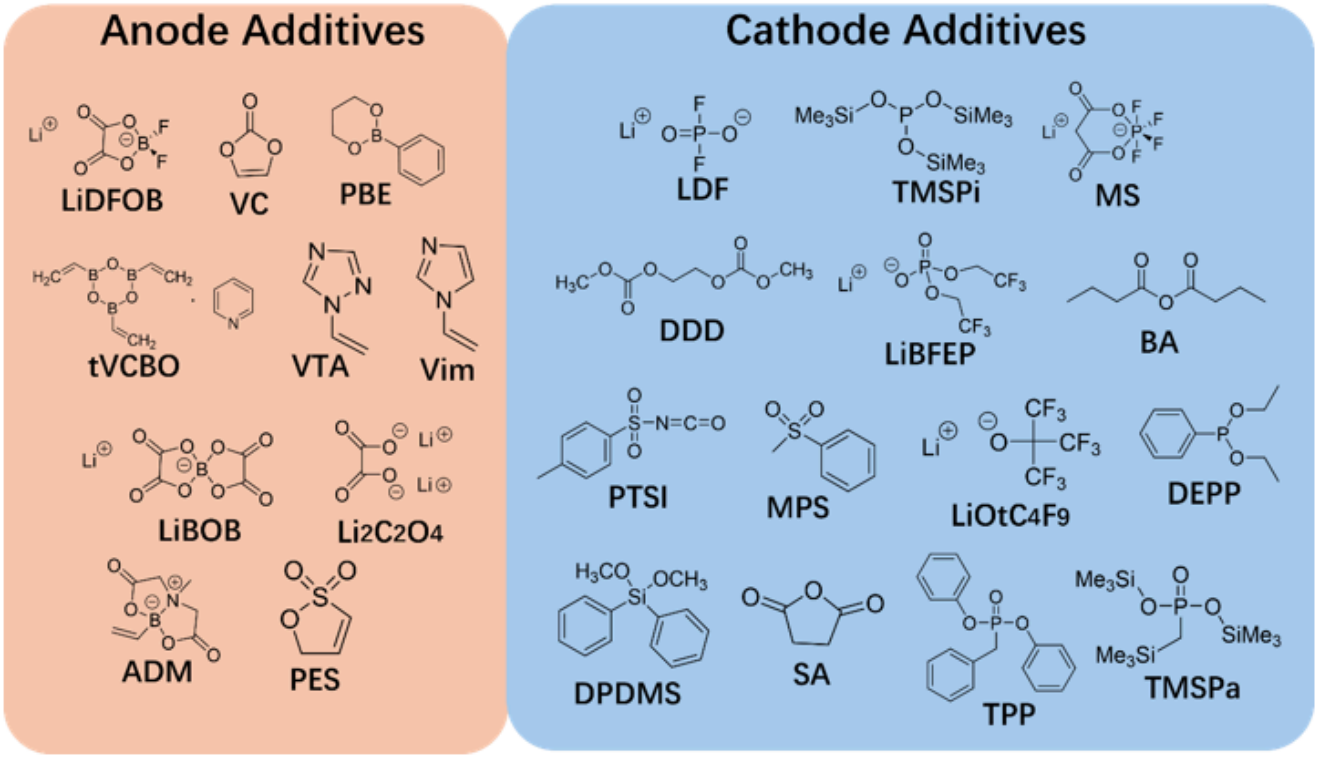

Figure 2

Molecular structures of the anode and cathode additives.[24, 25, 26, 27, 28, 29, 30, 31, 32] Their chemical names and acronyms are listed in SI.

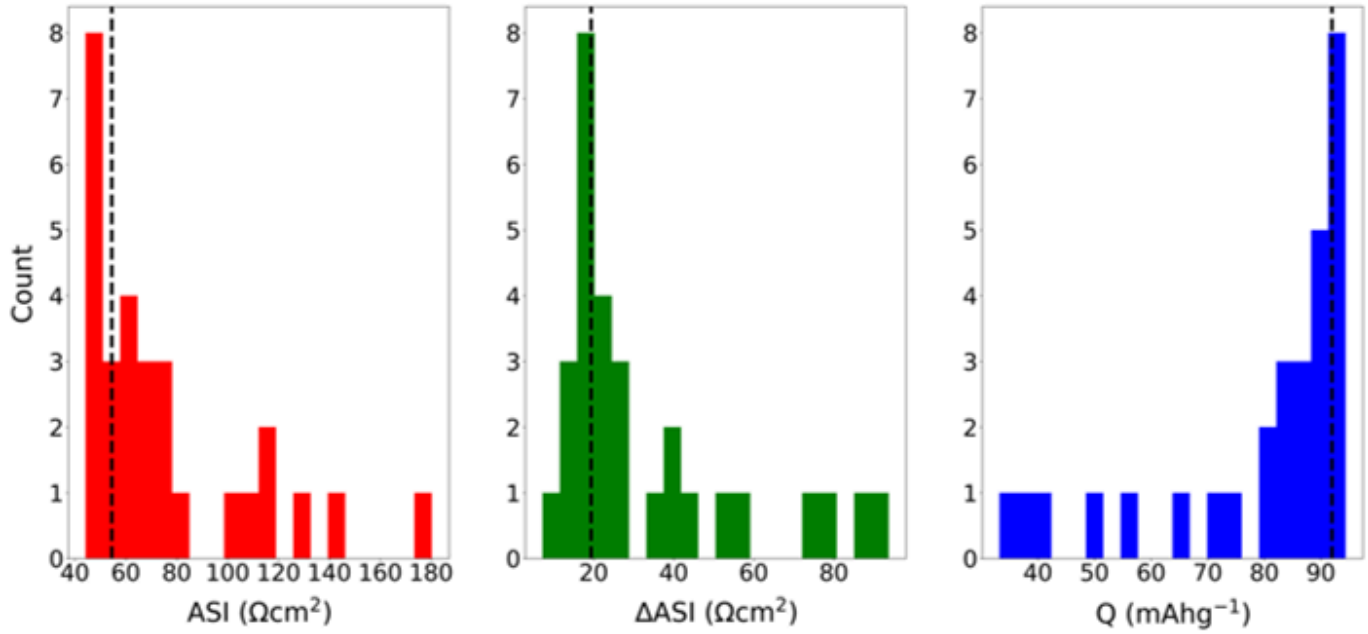

Figure 3

Distributions of measured final area specific impedance (ASI), impedance rise (ΔASI), and final specific capacity (Q) of the initial dataset of 28 additives and the baseline electrolyte. The black dashed lines indicate the measured values of the baseline solvent.

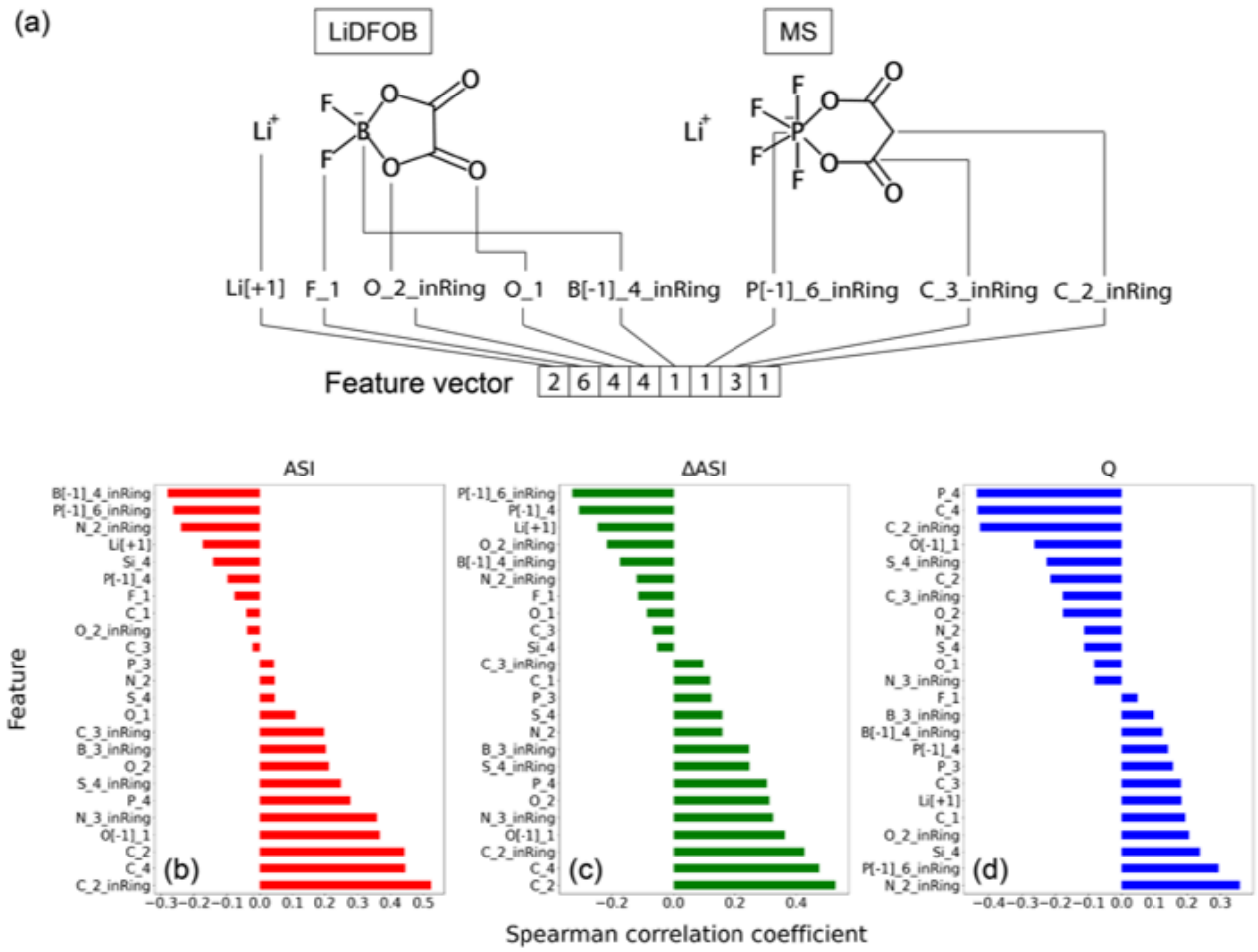

Figure 4

(a) Feature generation scheme used for molecular additives in the work. The number in the square brackets, after the underscore, and 'inRing' indicates the formal charge, the number of bonded neighbors (not including H), and whether the atom is part of a ring, respectively. Spearman correlation analysis of various features with respect to final area specific impedance (ASI), (b) impedance rise (ΔASI), and (c) final specific capacity (Q).

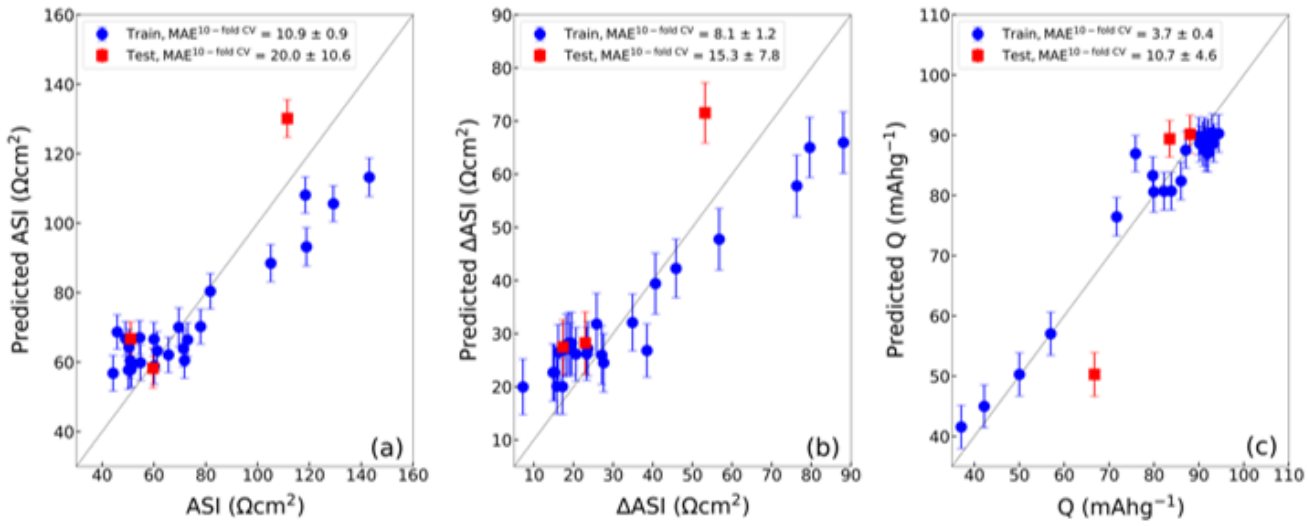

Figure 5

Parity plot of GPR-predicted (a) final area specific impedance (ASI), (b) impedance rise (ΔASI), and (c) final specific capacity (Q) versus measured values for the training set of 28 additive and baseline solvent. The mean absolute values (MAEs) based on 10-fold cross validation are shown in the legends. The error bars indicate one standard deviation.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SILNMOML3.3.docx
- LNMOwSingleadditivecustomFeatsHighlighted.xlsx
- FeaturetableforSI.xlsx
- TOC.png